

ASSIGNMENT 4

USING DAX AND VISUALIZATION

CALCULATED COLUMNS

STEP 1 :

Create a Calculated Column for 'Category Type'

Dax used

Category Type = 'Order Details'[Category]&"_"&'Order Details'[Sub-Category]

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type
B-25602	561	212	3	Clothing	Saree	Clothing_Saree
B-25602	119	-5	8	Clothing	Saree	Clothing_Saree
B-25603	193	-166	3	Clothing	Saree	Clothing_Saree
B-25604	157	5	9	Clothing	Saree	Clothing_Saree
B-25605	75	0	7	Clothing	Saree	Clothing_Saree
B-25609	25	-5	4	Clothing	Saree	Clothing_Saree
B-25610	43	0	3	Clothing	Saree	Clothing_Saree
B-25611	160	-59	2	Clothing	Saree	Clothing_Saree
B-25613	1603	0	9	Clothing	Saree	Clothing_Saree
B-25619	353	90	8	Clothing	Saree	Clothing_Saree
B-25622	534	0	3	Clothing	Saree	Clothing_Saree
B-25623	149	-87	4	Clothing	Saree	Clothing_Saree
B-25625	635	-349	5	Clothing	Saree	Clothing_Saree
B-25628	24	-9	4	Clothing	Saree	Clothing_Saree
B-25633	711	-8	4	Clothing	Saree	Clothing_Saree
B-25635	382	30	3	Clothing	Saree	Clothing_Saree
B-25636	637	113	5	Clothing	Saree	Clothing_Saree
B-25640	122	-47	4	Clothing	Saree	Clothing_Saree
B-25646	20	-8	2	Clothing	Saree	Clothing_Saree
B-25647	42	-6	4	Clothing	Saree	Clothing_Saree
B-25648	55	-26	4	Clothing	Saree	Clothing_Saree
B-25648	130	-41	4	Clothing	Saree	Clothing_Saree
B-25650	211	-105	2	Clothing	Saree	Clothing_Saree
B-25650	31	-2	2	Clothing	Saree	Clothing_Saree
B-25650	512	-225	5	Clothing	Saree	Clothing_Saree
B-25650	238	20	2	Clothing	Saree	Clothing_Saree
B-25651	23	4	1	Clothing	Saree	Clothing_Saree

STEP 2 :

Calculate Revenue per Order in Order Details Table

Dax Used

Revenue = 'Order Details'[Amount]*'Order Details'[Quantity]

File Home Help Table tools Column tools Share

Name Revenue \$% Format Whole number Summarization Sum Data type Whole number \$ % 0 Data category Uncategorized Sort by column Data groups Manage relationships New column Calculations

Structure Formatting Properties

1 Revenue = 'Order Details'[Amount]*'Order Details'[Quantity]

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue
B-25602	561	212	3	Clothing	Saree	Clothing_Saree	1683
B-25602	119	-5	8	Clothing	Saree	Clothing_Saree	952
B-25603	193	-166	3	Clothing	Saree	Clothing_Saree	579
B-25604	157	5	9	Clothing	Saree	Clothing_Saree	1413
B-25605	75	0	7	Clothing	Saree	Clothing_Saree	525
B-25609	25	-5	4	Clothing	Saree	Clothing_Saree	100
B-25610	43	0	3	Clothing	Saree	Clothing_Saree	129
B-25611	160	-59	2	Clothing	Saree	Clothing_Saree	320
B-25613	1603	0	9	Clothing	Saree	Clothing_Saree	14427
B-25619	353	90	8	Clothing	Saree	Clothing_Saree	2824
B-25622	534	0	3	Clothing	Saree	Clothing_Saree	1602
B-25623	149	-87	4	Clothing	Saree	Clothing_Saree	596
B-25625	635	-349	5	Clothing	Saree	Clothing_Saree	3175
B-25628	24	-9	4	Clothing	Saree	Clothing_Saree	96
B-25633	711	-8	4	Clothing	Saree	Clothing_Saree	2844
B-25635	382	30	3	Clothing	Saree	Clothing_Saree	1146
B-25636	637	113	5	Clothing	Saree	Clothing_Saree	3185
B-25640	122	-47	4	Clothing	Saree	Clothing_Saree	488
B-25646	20	-8	2	Clothing	Saree	Clothing_Saree	40
B-25647	42	-6	4	Clothing	Saree	Clothing_Saree	168
B-25648	55	-26	4	Clothing	Saree	Clothing_Saree	220
B-25648	130	-41	4	Clothing	Saree	Clothing_Saree	520
B-25650	211	-105	2	Clothing	Saree	Clothing_Saree	422
B-25650	31	-2	2	Clothing	Saree	Clothing_Saree	62
B-25650	512	-225	5	Clothing	Saree	Clothing_Saree	2560
B-25650	238	20	2	Clothing	Saree	Clothing_Saree	476
B-25651	23	4	1	Clothing	Saree	Clothing_Saree	23
B-25651	467	41	4	Clothing	Saree	Clothing_Saree	1868

Table: Order Details (1,500 rows) Columns: Revenue (845 distinct values)

Data

Search

- List of Orders
 - City
 - CustomerName
- Order Date
 - Date Hierarchy
 - Order ID
 - State
- Order Details
 - Amount
 - Category
 - Category Type
 - Order ID
 - Profit
 - Quantity
 - Revenue
 - Sub-Category
- Sales target
 - Category
 - Month of Order Date
 - Date Hierarchy
 - Target

STEP 3:

Create a Calculated Column to Categorize Sales

Dax Used

Sales Category = IF('Order Details'[Amount]>AVERAGE('Order Details'[Amount]),"Above Average","Below Average")

File Home Help Table tools Column tools Share

Name Sales Category \$% Format Text Summarization Don't summarize Data type Text \$ % Auto Data category Uncategorized Sort by column Data groups Manage relationships New column Calculations

Structure Formatting Properties

1 Sales Category = IF('Order Details'[Amount]>AVERAGE('Order Details'[Amount]),"Above Average","Below Average")

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue	Sales Category
B-25602	561	212	3	Clothing	Saree	Clothing_Saree	1683	Above Average
B-25602	119	-5	8	Clothing	Saree	Clothing_Saree	952	Below Average
B-25603	193	-166	3	Clothing	Saree	Clothing_Saree	579	Below Average
B-25604	157	5	9	Clothing	Saree	Clothing_Saree	1413	Below Average
B-25605	75	0	7	Clothing	Saree	Clothing_Saree	525	Below Average
B-25609	25	-5	4	Clothing	Saree	Clothing_Saree	100	Below Average
B-25610	43	0	3	Clothing	Saree	Clothing_Saree	129	Below Average
B-25611	160	-59	2	Clothing	Saree	Clothing_Saree	320	Below Average
B-25613	1603	0	9	Clothing	Saree	Clothing_Saree	14427	Above Average
B-25619	353	90	8	Clothing	Saree	Clothing_Saree	2824	Above Average
B-25622	534	0	3	Clothing	Saree	Clothing_Saree	1602	Above Average
B-25623	149	-87	4	Clothing	Saree	Clothing_Saree	596	Below Average
B-25625	635	-349	5	Clothing	Saree	Clothing_Saree	3175	Above Average
B-25628	24	-9	4	Clothing	Saree	Clothing_Saree	96	Below Average
B-25633	711	-8	4	Clothing	Saree	Clothing_Saree	2844	Above Average
B-25635	382	30	3	Clothing	Saree	Clothing_Saree	1146	Above Average
B-25636	637	113	5	Clothing	Saree	Clothing_Saree	3185	Above Average
B-25640	122	-47	4	Clothing	Saree	Clothing_Saree	488	Below Average
B-25646	20	-8	2	Clothing	Saree	Clothing_Saree	40	Below Average
B-25647	42	-6	4	Clothing	Saree	Clothing_Saree	168	Below Average
B-25648	55	-26	4	Clothing	Saree	Clothing_Saree	220	Below Average
B-25648	130	-41	4	Clothing	Saree	Clothing_Saree	520	Below Average
B-25650	211	-105	2	Clothing	Saree	Clothing_Saree	422	Below Average
B-25650	31	-2	2	Clothing	Saree	Clothing_Saree	62	Below Average
B-25650	512	-225	5	Clothing	Saree	Clothing_Saree	2560	Above Average
B-25650	238	20	2	Clothing	Saree	Clothing_Saree	476	Below Average
B-25651	23	4	1	Clothing	Saree	Clothing_Saree	23	Below Average
B-25651	467	41	4	Clothing	Saree	Clothing_Saree	1868	Above Average

Table: Order Details (1,500 rows) Columns: Sales Category (2 distinct values)

Data

Search

- List of Orders
 - City
 - CustomerName
- Order Date
 - Date Hierarchy
 - Order ID
 - State
- Order Details
 - Amount
 - Category
 - Category Type
 - Order ID
 - Profit
 - Quantity
 - Revenue
 - Sales Category
 - Sub-Category
- Sales target
 - Category
 - Month of Order Date
 - Date Hierarchy
 - Target

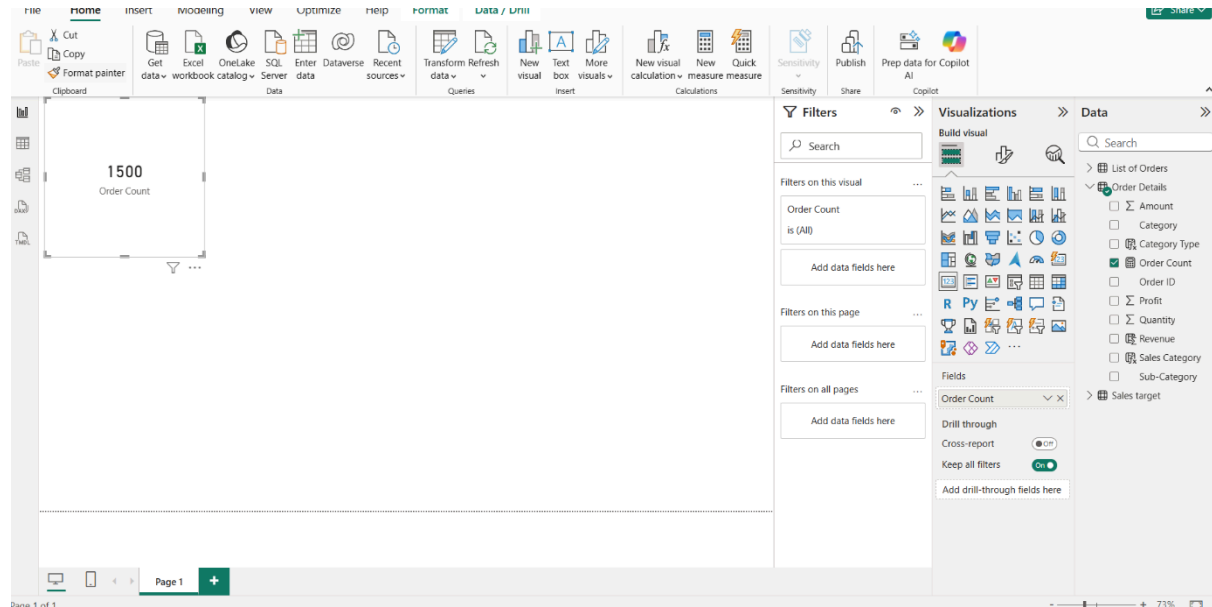
CALCULATED MEASURE

STEP 1:

Calculate Order Count

Dax Used

Order Count = COUNT('Order Details'[Order ID])

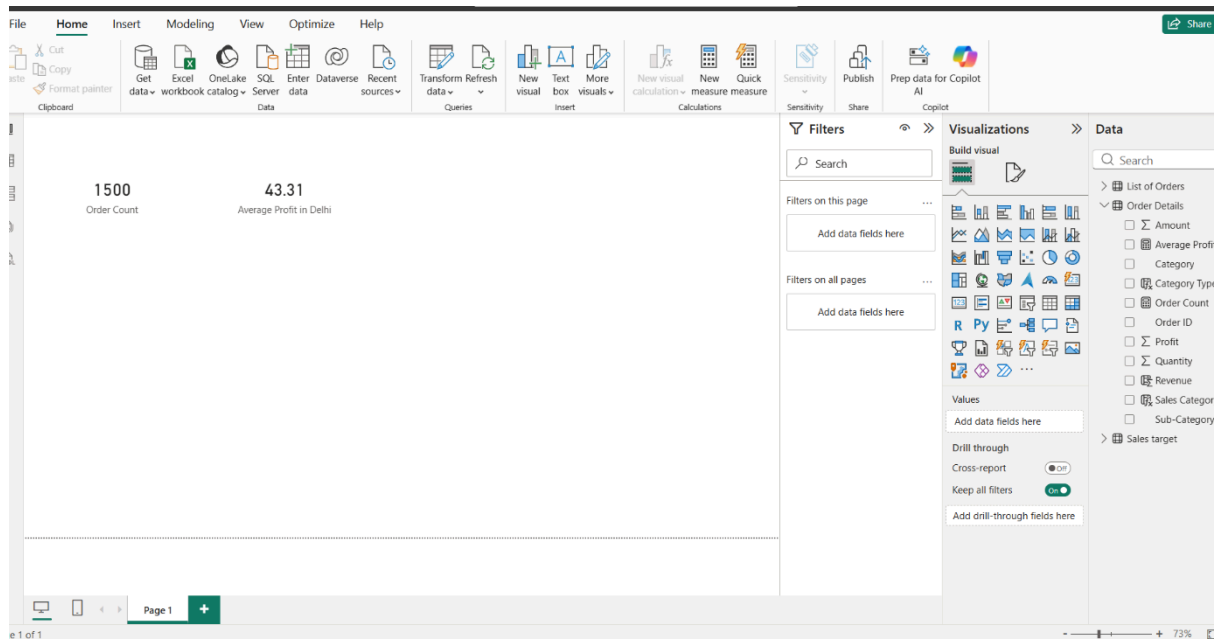


STEP 2:

Calculate Average Profit in Delhi

Dax Used

Average Profit in Delhi = CALCULATE(AVERAGE('Order Details'[Profit]),'List of Orders'[City]="Delhi")

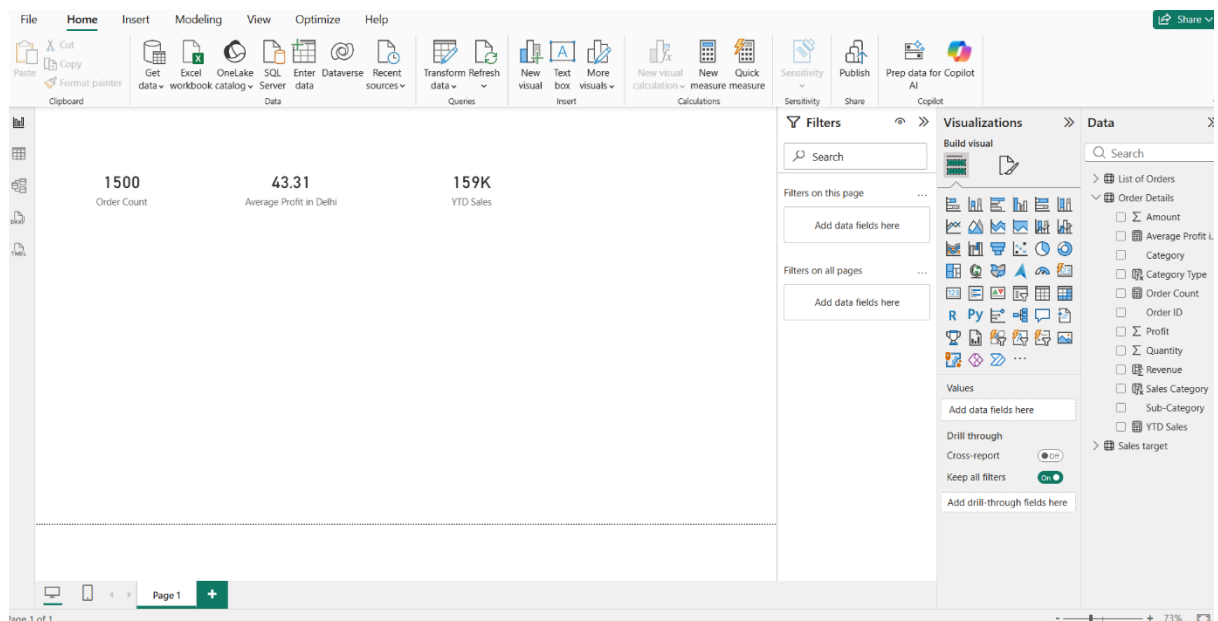


STEP 3:

Calculate Year-to-Date (YTD) Sales

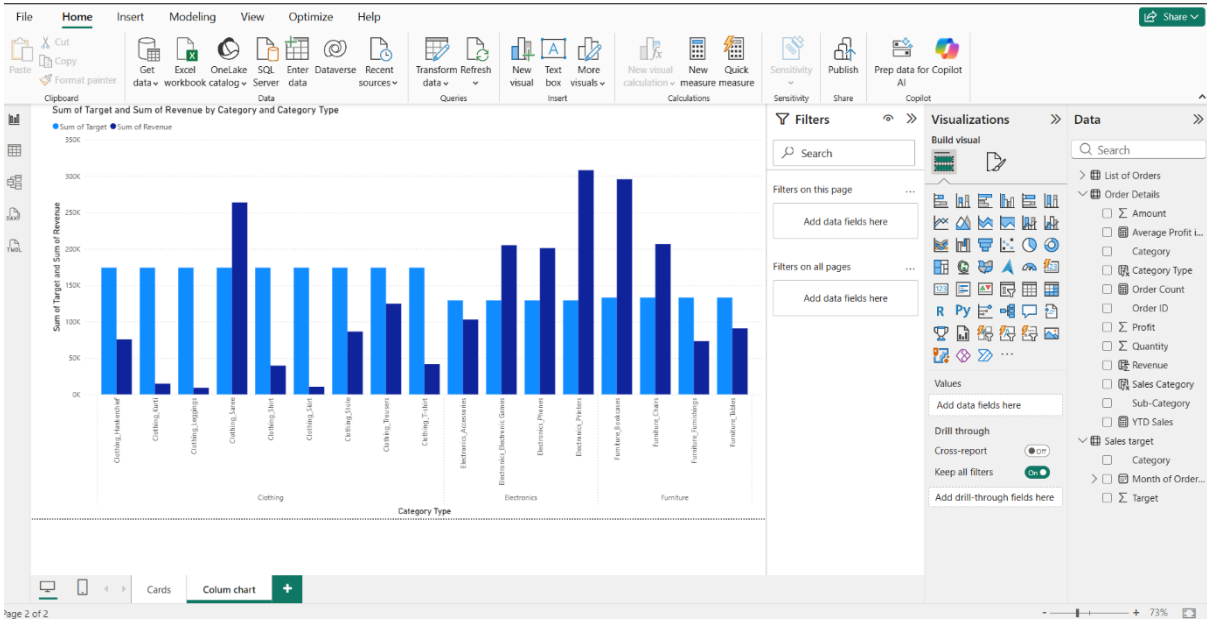
Dax Used

YTD Sales = TOTALYTD(SUM('Order Details'[Amount]),'List of Orders'[Order Date])

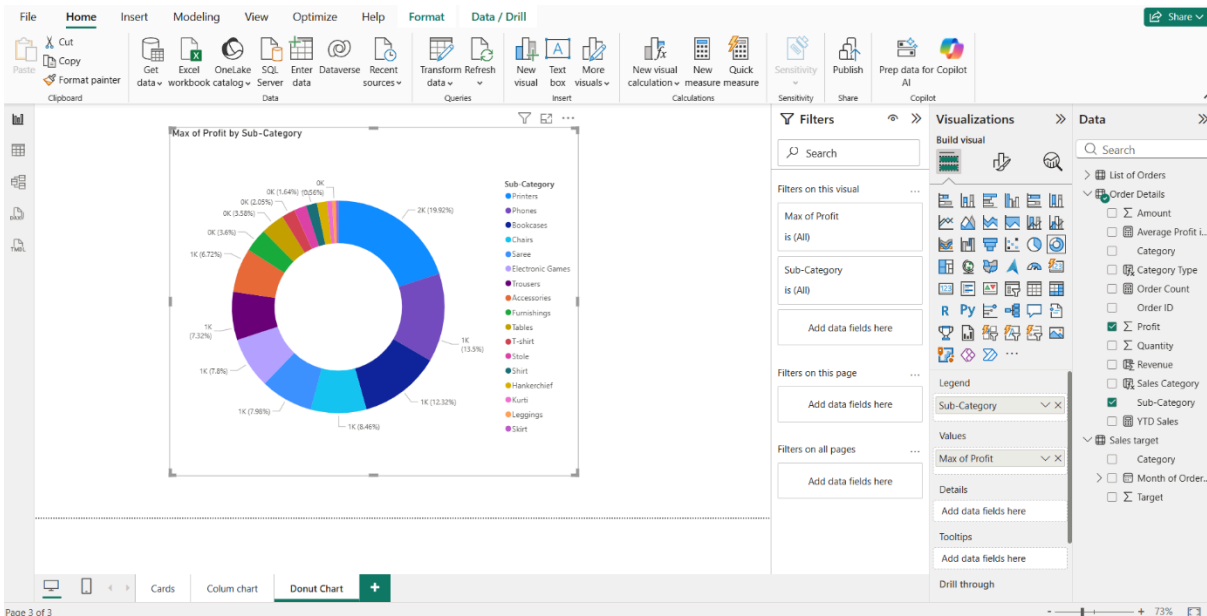


DATA VISUALIZATION

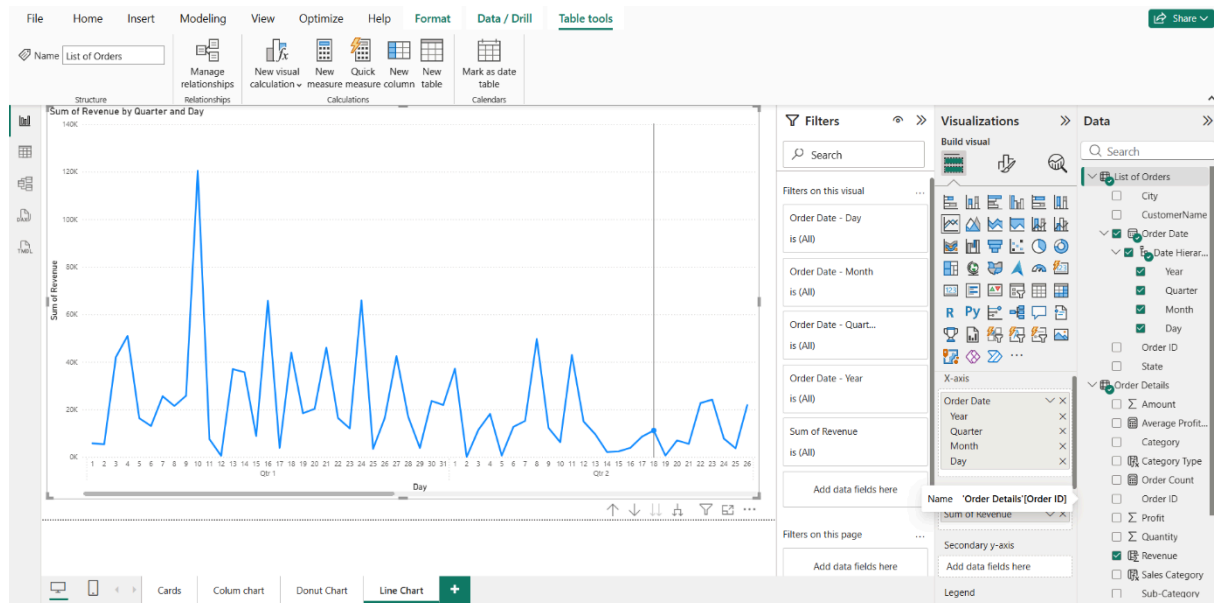
Sales Target Achievement by Category



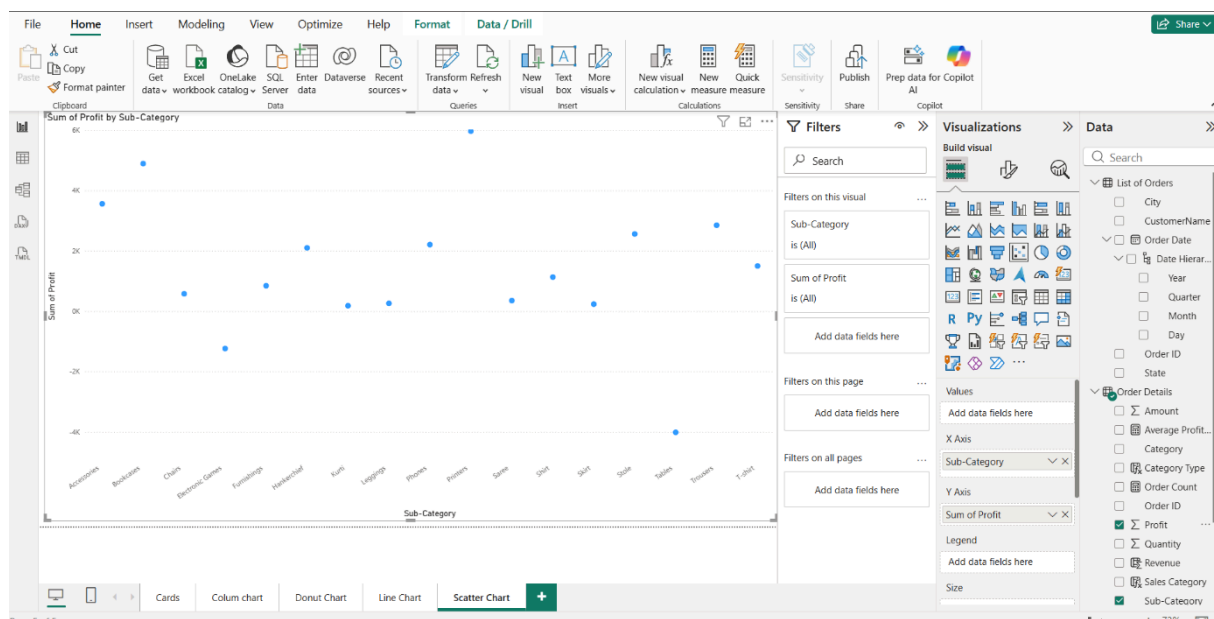
Max Profit Margin by Sub-Category



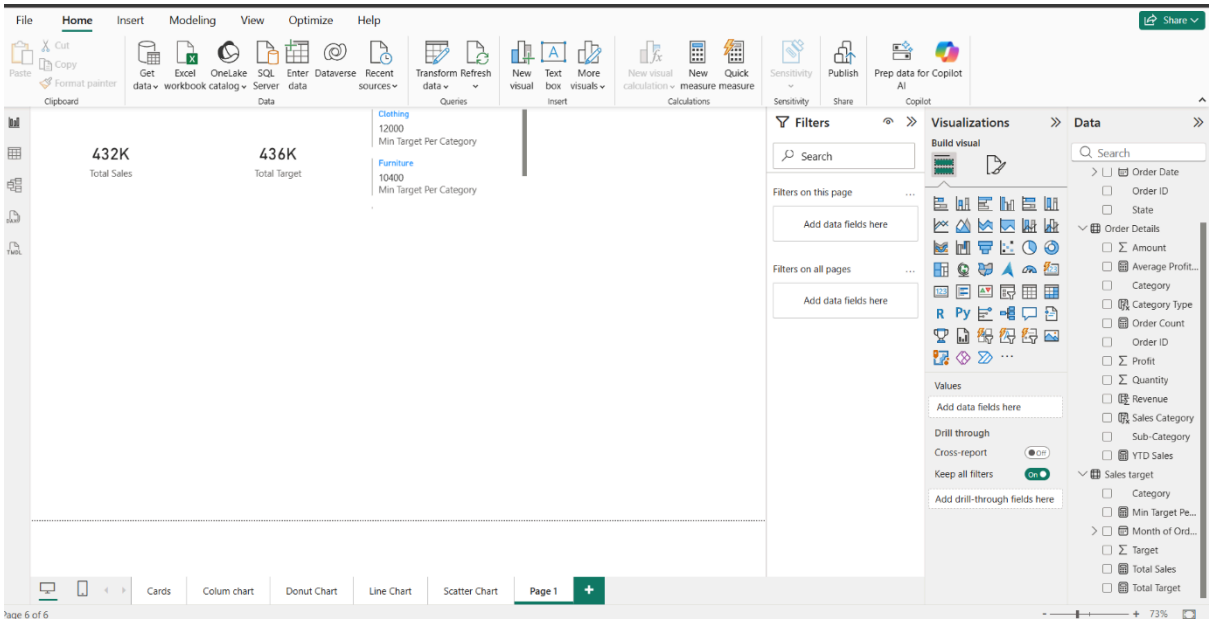
Monthly Sales Trend



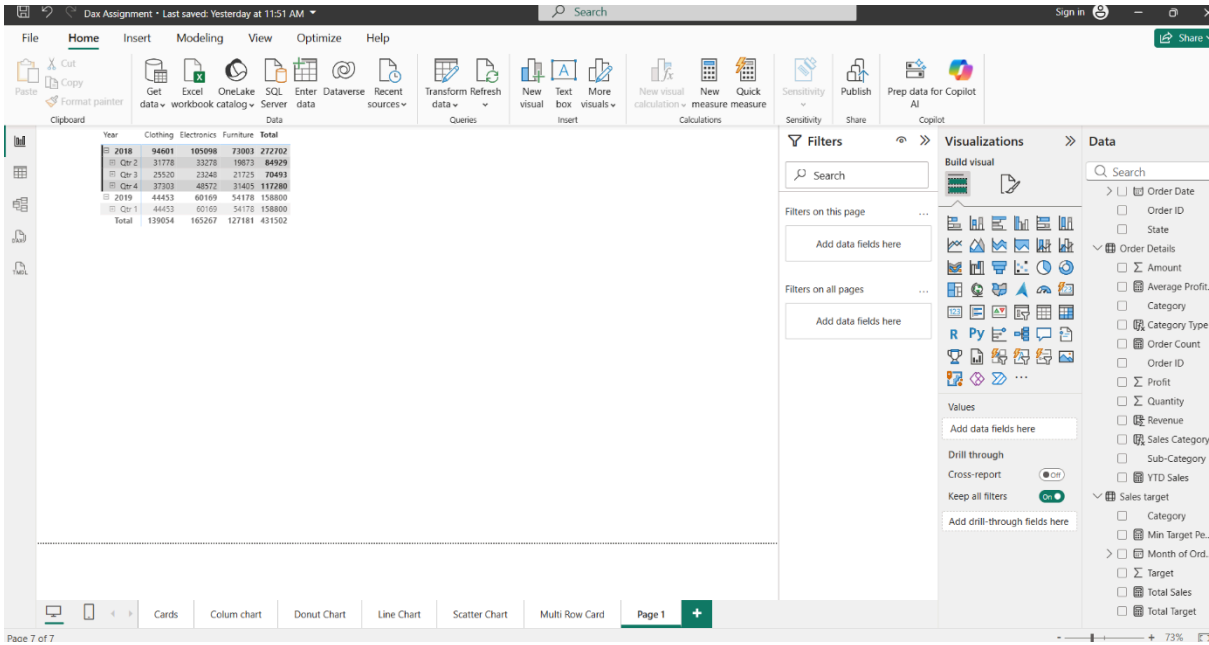
Comparison of Profit and Quantity by Sub-Category



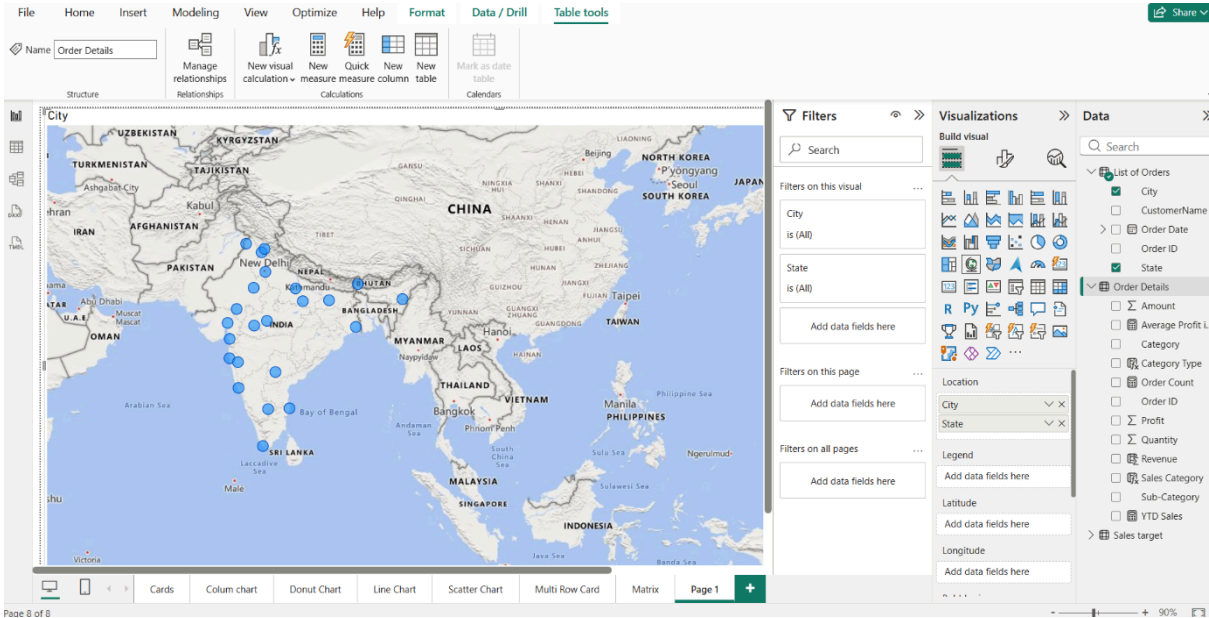
Comparison of Total Sales Amount and Target



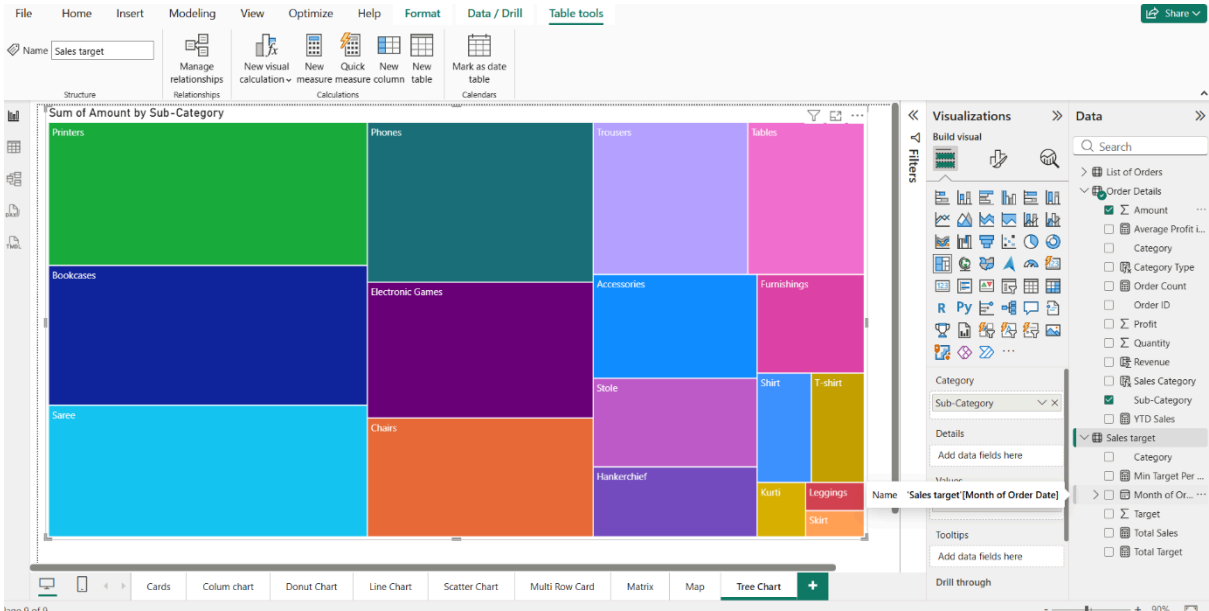
Sales Performance Matrix



Geographic Sales Analysis



Sales Distribution by Sub-Category



Order Count Analysis by State

